

Statistics: Preliminary Ph.D. Course

Day 3: The Measure-Theoretic Approach to Probability

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Today's Session

Formalizing what we have seen

- Why **Measure Theory**: Limitations of the Riemann Integral. The Dirichlet Function.
- **Measurable Functions** and **Measures**. Dirac, Counting, Lebesgue, and Probability Measures. Definitions, Theorems, Proofs, and Examples.
- Definition of the **Abstract Integral**. From Indicators to Simple Functions and to Measurable Functions. The **Lebesgue Integral** and the **Expectation**. Examples.

If time permits...

- The **Radon-Nikodym Derivative**. Motivation and Theorem.
- Induced Law and **Distribution**.
- **Change of Variables** Formula. Theorem and Proof.
- **PMFs** and **PDFs** as Radon-Nikodým derivatives.

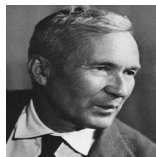
Relation to Measure Theory

- Due to the technicality we mentioned on the first day, we will introduce the approach in **Measure Theory**. This will also enable the unification of both the continuous and discrete cases.
- For that, we need to introduce the **Lebesgue Integral**. It is more general than the **Riemann Integral**, which is the typical one from high school. Hence, we will motivate the new integral based on the limitations of the Riemann integral.
- Keep in mind that an integral is not simply finding an antiderivative. Think about integrals of functions over sets as measures of those functions... More on this now.
- For simplicity, we will focus on the **univariate case**, but everything could be readily extended to higher dimensions.

Motivation

A Bit of History

Limitations of Riemann Integral $\xRightarrow{\text{Lebesgue+Borel}}$ Measure Theory $\xRightarrow{\text{Kolmogorov}}$ Modern Probability



- **Problem:** What is the probability that a number chosen uniformly at random in the unit interval is a rational number?
- Formally, the question is asking to compute the following probability:

$$\mathbb{P}(X \in \mathbb{Q}) = \int_{\mathbb{Q}} f(x) \, dx = \int_{\mathbb{Q}} \mathbb{1}_{\{0 < x < 1\}} \, dx = \int_{\mathbb{Q} \cap (0,1)} dx = \int_0^1 \mathbb{1}_{\mathbb{Q}}(x) \, dx$$

Motivation

Limitations of the Riemann Integral

- **Problem:** What is the probability that a number chosen uniformly at random in the unit interval is a rational number?

$$\mathbb{P}(X \in \mathbb{Q}) = \int_0^1 \mathbb{1}_{\mathbb{Q}}(x) \, dx$$

- $\mathbb{1}_{\mathbb{Q}}(x) = \mathbb{1}_{\{x \in \mathbb{Q}\}}$ is the **Dirichlet Function**. It takes value 1 if x is a rational number and 0 otherwise.
- How do we compute this integral? Can we use the Riemann integral?
- Let's go through the basics of the Riemann Integral.

The Riemann Integral

Upper and Lower Sums

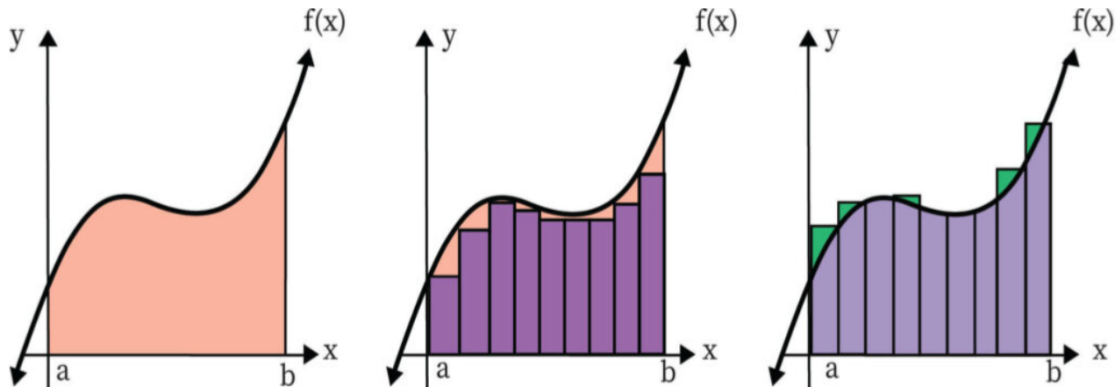
- Let f be a non-negative real-valued and bounded function on the interval $[a, b]$, i.e.,
 $f : [a, b] \longrightarrow \mathbb{R}$.
 $x \mapsto y$
- Define a **partition** P of the interval $[a, b]$ as a finite sequence of numbers of the form:
 $P = \{a = x_0 < x_1 < \cdots < x_i < \cdots < x_n = b\}$. Each $[x_{i-1}, x_i] \quad \forall i \in \{1, \dots, n\}$ is called a **sub-interval** of the partition.
- For each $i = 1, \dots, n$, let: $m_i = \inf_{x \in [x_{i-1}, x_i]} f(x)$ and $M_i = \sup_{x \in [x_{i-1}, x_i]} f(x)$
- Define the **lower** (Darboux) **sum** $L(P, f)$ and the **upper** (Darboux) **sum** $U(P, f)$ as:

$$L(P, f) = \sum_{i=1}^n m_i(x_i - x_{i-1}) \quad U(P, f) = \sum_{i=1}^n M_i(x_i - x_{i-1})$$

Note that each term in the sum is the area of a rectangle.

The Riemann Integral

Upper and Lower Sums



Motivation

The Riemann Integral

- Define the **lower integral** $I_*(f)$ and **upper integral** $I^*(f)$ on $[a, b]$ as:

$$I_*(f) = \sup_P L(P, f) \quad I^*(f) = \inf_P U(P, f)$$

Theorem 3.1 (Riemann Integrability)

Let f be a non-negative, real-valued, and bounded function on the interval $[a, b]$. f is Riemann Integrable if and only if

$$I(f) = \int_a^b f(x) \, dx = I_*(f) = I^*(f)$$

i.e., if the upper and lower integrals are the same for any choice of partition P . $I(f)$ is the area under f in the interval $[a, b]$.

Nice illustration. We can now address the initial problem.

Motivation

Limitations of the Riemann Integral

- Usually the definition is difficult to apply (any P !), so we use the **Fundamental Theorem of Calculus** and compute antiderivatives. This makes integration way easier!
- Remember our problem: $\mathbb{P}(X \in \mathbb{Q}) = \int_0^1 \mathbb{1}_{\mathbb{Q}}(x) dx$. Can we use the Riemann integral?

Answer (The Dirichlet Function is not Riemann Integrable)

$$\int_0^1 \mathbb{1}_{\mathbb{Q}}(x) dx = \nexists$$

Proof. (Hint: Apply the definition of Riemann Integrability by taking an arbitrary partition P using the fact that $\mathbb{Q}$ is dense in $\mathbb{R}$).

- Can we still compute the probability? Yes, but we need another integral: the **Lebesgue Integral**.
- For this task, we need to introduce more mathematical machinery.

Relation to Measure Theory

Measurable Functions

- **Definition 3.1:** Let $f : \Omega \rightarrow B$ be a function. Let $D \subseteq B$. The **inverse image** or **pre-image** of f as the set $f^{-1}(D) := \{\omega \in \Omega : f(\omega) \in D\}$. *Example.*
- **Claim 3.1:** Let $D_1, \dots, D_n$ be subsets of B . Then:
 1. $f^{-1}(D^c) = (f^{-1}(D))^c$
 2. $f^{-1}(\cup_n D_n) = \cup_n f^{-1}(D_n)$

Proof.

- **Definition 3.2:** Let $(\Omega, \mathcal{F})$ and $(\Lambda, \mathcal{G})$ be measurable spaces and let $f(\omega)$ be a function $f : \Omega \rightarrow \Lambda$. We say that f is a **measurable function** from $(\Omega, \mathcal{F})$ to $(\Lambda, \mathcal{G})$ if

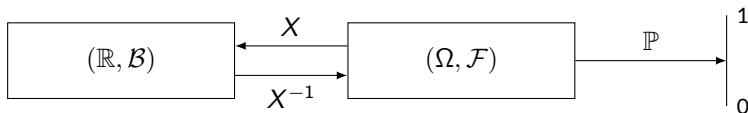
$$f^{-1}(G) = \{\omega \in \Omega : f(\omega) \in G\} \in \mathcal{F}, \quad \forall G \in \mathcal{G}$$

Examples.

Relation to Measure Theory

Measurable Functions

- If $\Lambda = \mathbb{R}$ and $\mathcal{G} = \mathcal{B}$, $f(\omega)$ is said to be **Borel-measurable** (Borel function).
- **Claim 3.2:** Let f be measurable from $(\Omega, \mathcal{F})$ to $(\Lambda, \mathcal{G})$. Then $f^{-1}(\mathcal{G}) := \{f^{-1}(G) : G \in \mathcal{G}\}$ is a sub- σ -field of $\mathcal{F}$. **Proof.**
- **Claim 3.3:** The function $f : \Omega \rightarrow \Lambda$ is measurable from $(\Omega, \mathcal{P}(\Omega))$ to $(\Lambda, \mathcal{G})$. **Proof.**
- If $X(\omega)$ is measurable from $(\Omega, \mathcal{F})$ to $(\mathbb{R}, \mathcal{B})$ then $X(\omega)$ is a **random variable**.



- **Exercise:** Prove that the rational numbers are in the Borel σ -algebra, i.e., $\mathbb{Q} \in \mathcal{B}$. Use the definition of $\mathcal{B}$ from Day 1 and use the fact that $\mathbb{Q}$ is countable. Is the Dirichlet function measurable?

Relation to Measure Theory

Measures

- **Definition 3.2:** Let $(\Omega, \mathcal{F})$ be a measurable space. A set function $\mu : \mathcal{F} \rightarrow [0, \infty]$ is a **measure** if it satisfies
 1. $\mu(\emptyset) = 0$
 2. If $A_1, A_2, \dots \in \mathcal{F}$ and $A_i \cap A_j = \emptyset$ for $i \neq j$, then $\mu(\cup_n A_n) = \sum_n \mu(A_n)$.
- **Example 3.1:** Let $(\Omega, \mathcal{F})$ be a measurable space and $A \in \mathcal{F}$. The **Dirac measure** or point mass at c is given by

$$\delta_c(A) = \begin{cases} 0, & c \notin A \\ 1, & c \in A \end{cases}$$

- **Example 3.2:** Let $(\Omega, \mathcal{F})$ be a measurable space and let $\nu(A)$ be the number of elements in $A \in \mathcal{F}$ ($\nu(A) = \infty$ if A contains infinitely many elements). Then, ν is the **counting measure** on $\mathcal{F}$. It is also the countable sum of Dirac measures: $\sum_{c \in A} \delta_c(A)$.

Relation to Measure Theory

Measures

- **Example 3.3:** There is a unique measure λ on $(\mathbb{R}, \mathcal{B})$ that satisfies $\lambda([a, b]) = b - a$ for any finite interval $-\infty < a \leq b < \infty$. This measure λ is the **Lebesgue measure**.
- **Definition 3.6:** If a measure $\mathbb{P}$ satisfies $\mathbb{P}(\Omega) = 1$, it is called **probability measure**. Equivalently, if we restrict λ to the measurable space $([0, 1], \mathcal{B}_{[0,1]})$, the resulting measure $\mathbb{P}$ is the probability measure.
- The triplet $(\Omega, \mathcal{F}, \mu)$ is a **measure space**. Does the definition of measure ring a bell? Recall the Axioms on $\mathbb{P}$.

Relation to Measure Theory

Integration

- **Definition 3.3:** A simple function $\phi : \Omega \rightarrow \mathbb{R}$ is a linear combination of indicators $\mathbb{1}_A(\omega)$ of measurable sets $A_1, A_2, \dots \in \mathcal{F}$, i.e.,

$$\phi(\omega) = \sum_{i=1}^n a_i \mathbb{1}_{A_i}(\omega)$$

where $a_1, a_2, \dots \in \mathbb{R}$ and $A_1, A_2, \dots \subseteq \Omega$.

- **Definition 3.6:** The **positive part of a function** f is given by $f^+(\omega) := \max\{f(\omega), 0\}$. The **negative part of a function** f is given by $f^-(\omega) := \max\{-f(\omega), 0\}$.

Theorem 3.2 (Pointwise Limit of Simple Functions)

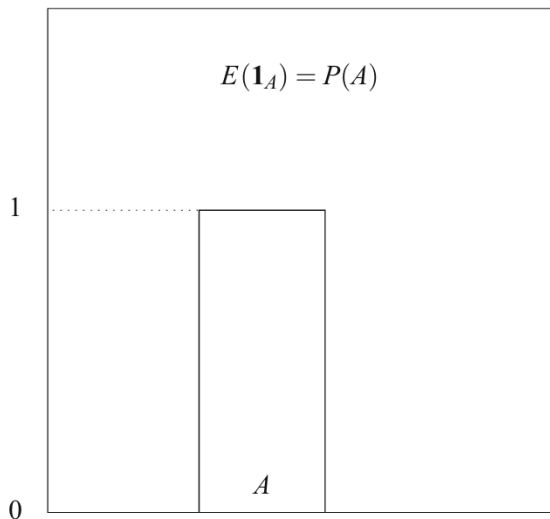
Let $(\Omega, \mathcal{F})$ be a measurable space and let $f : \Omega \rightarrow \mathbb{R}_+$ be a positive $\mathcal{F}$ -measurable function. Then, there exists a non-decreasing sequence $\{\phi_n\}_{n \in \mathbb{N}}$ of simple functions such that $\forall \omega \in \Omega$

$$\lim_{n \rightarrow \infty} \phi_n(\omega) = f(\omega).$$

Relation to Measure Theory

Integration

We now introduce the construction of the **abstract integral**, i.e., w.r.t. any measure μ .



Definition 3.7: (Four Steps)

– Step 1: Indicator Functions

$$\mathbb{1}_A = \mathbb{1}_{\{\omega \in A\}}$$

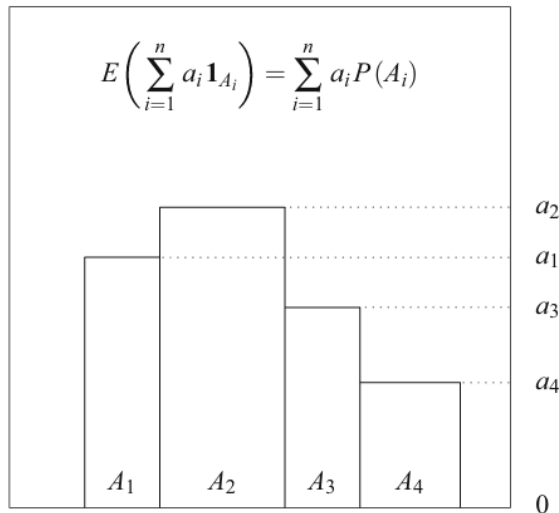
Think about integrals as measures of functions. It is natural to think about the measure of the indicator as the area of the rectangle: *height* $\times$ *length*. The height is 1, but A can be very abstract, so we use its measure instead. Hence:

$$\int_{\Omega} \mathbb{1}_A d\mu = \mu(A)$$

Relation to Measure Theory

Integration

We now complicate a bit more the function to be integrated.



- Step 2: Simple Functions $\phi : \Omega \rightarrow \mathbb{R}$ with finite range.

$$\phi(\omega) = \sum_i a_i \mathbb{1}_{A_i}(\omega)$$

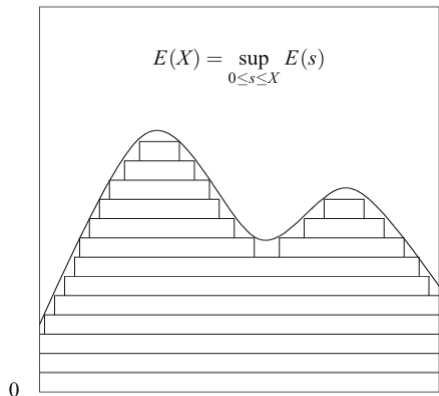
They are just linear combinations of indicators. We can then add up the areas of the different rectangles. Hence:

$$\begin{aligned} \int_{\Omega} \phi(\omega) d\mu &= \int_{\Omega} \sum_i a_i \mathbb{1}_{A_i} d\mu \\ &= \sum_i a_i \int_{\Omega} \mathbb{1}_{A_i} d\mu = \sum_i a_i \mu(A_i) \end{aligned}$$

Relation to Measure Theory

Integration

And another layer of complication...



– Step 3: Positive Measurable Functions

$$X(\omega) \geq 0$$

Let $\mathcal{S}_X$ be the collection of all non-negative simple functions such that $\phi(\omega) \leq X(\omega) \forall \omega \in \Omega$. The integral of X with respect to μ is defined as:

$$\int_{\Omega} X d\mu = \sup \left\{ \int \phi d\mu : \phi \in \mathcal{S}_X \right\}$$

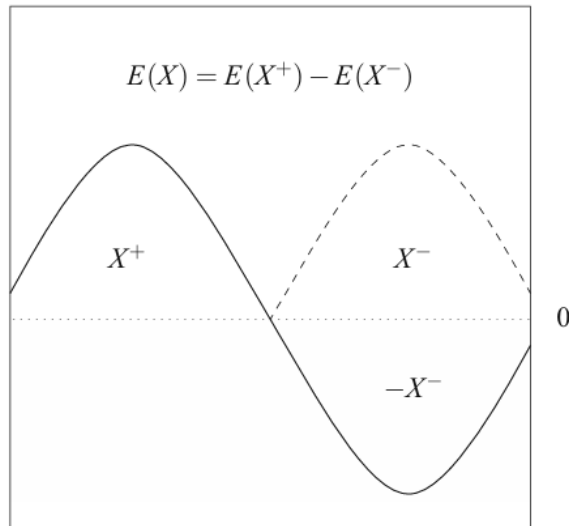
Hence, for all X there exists a sequence

of simple functions $\phi_1, \phi_2, \dots$ such that $0 \leq \phi_i \leq X \forall i$ and $\lim_{n \rightarrow \infty} \int \phi_n d\mu = \int X d\mu$

Relation to Measure Theory

Integration

And finally:



- Step 4: Integrable Functions They are much more general than Riemann integrable functions. This one is easy, we just integrate the positive and negative parts separately:

$$\int_{\Omega} X \, d\mu = \int_{\Omega} X^+ \, d\mu - \int_{\Omega} X^- \, d\mu$$

Problem: if both integrals are ∞ , we have an indetermination ($\infty \cdot 0 = 0$ by convention). Thus, we need:

$$\int_{\Omega} |X| \, d\mu < \infty \quad (\text{Integrability}) \quad \square$$

Relation to Measure Theory

Integration

- We can now define the **Lebesgue Integral**, i.e., the integral w.r.t. the Lebesgue Measure, λ :

$$\int_{\Omega} X \, d\lambda = \int_{\Omega} X(\omega) \, d\lambda(\omega) = \int_{\Omega} X(\omega) \, \lambda(d\omega)$$

All functions that are Riemann integrable are also Lebesgue integrable, in which case the Lebesgue integral is equal to the Riemann integral.

- Initial problem: $\mathbb{P}(X \in \mathbb{Q}) = \int_0^1 \mathbb{1}_{\mathbb{Q}}(x) \, dx = \nexists$. We have shown that the Dirichlet function is measurable. We could thus integrate it w.r.t. the Lebesgue measure.

Answer (Lebesgue Integral of the Dirichlet Function)

$$\int_0^1 \mathbb{1}_{\mathbb{Q}}(x) \, d\lambda = 0$$

Proof. (Hint: Use Integration Step 1, the definition of Lebesgue measure, and the fact that $\mathbb{Q}$ is countable so that it can be expressed as a countable union of disjoint singletons).

Relation to Measure Theory

Examples of Integration under some Important Measures

- **Claim 3.4** Let $(\Omega, \mathcal{F}, \delta_c)$ be a measure space and let $X : \Omega \rightarrow \mathbb{R}$ be a function. Then

$$\int_{\Omega} X(\omega) d\delta_c(\omega) = X(c)$$

Proof.

- **Claim 3.5** Let $(\Omega, \mathcal{F}, \nu)$ be a measure space where Ω is countable and $\mathcal{F} = \mathcal{P}(\Omega)$. Let $X : \Omega \rightarrow \mathbb{R}$ be a function. Then

$$\int_{\Omega} X(\omega) d\nu(\omega) = \sum_{\omega \in \Omega} X(\omega)$$

Proof.

- **Claim 3.6** If $\Omega = \mathbb{R}$ and $\mathcal{F} = \mathcal{B}$, then the integral of $X(\omega)$ w.r.t. λ over an interval $[a, b]$ agrees with the Riemann integral in calculus when the latter is well defined:

$$\int_{[a,b]} X(\omega) d\lambda(\omega) = \int_a^b X(\omega) d\omega$$

Relation to Measure Theory

Integration

We can now translate all this into the context of probability.

- Measurable Sets, $A \equiv$ Events.
- Measurable Functions, $X(\omega) \equiv$ Random Variables. $X(\omega) : (\Omega, \mathcal{F}, \mathbb{P}) \xrightarrow{\omega \mapsto X} (\mathbb{R}, \mathcal{B})$
- General Measure, $\mu \equiv$ Probability Measure, $\mathbb{P}$.
- Generic Measure Space, $(\Omega, \mathcal{F}, \mu) \equiv$ Probability Space $(\Omega, \mathcal{F}, \mathbb{P})$.
- **Definition 3.8:** The **Expectation** of a random variable X is the integral with respect to the Probability Measure, $\mathbb{P}$, in the measurable space $(\Omega, \mathcal{F})$ i.e.,

$$\mathbb{E}[X] = \int_{\Omega} X \, d\mathbb{P}$$

Problem: How do we compute the integral in Definition 3.8? It seems difficult to handle $\mathbb{P}$, right? It is. We need another important and abstract tool (yes, another one).

The rest of the story. If time permits...

Relation to Measure Theory

The Radon-Nikodým Derivative and the Change of Variables Formula

- To simplify the computation of an integral w.r.t. the probability measure we need the notion of **Distribution** of a random variable. Recall that a random variable is a measurable function:

$$X(\omega) : (\Omega, \mathcal{F}, \mathbb{P}) \longrightarrow (\mathbb{R}, \mathcal{B})$$

$\omega \mapsto X$

And the (mysterious) probability measure $\mathbb{P}$ is defined in $(\Omega, \mathcal{F})$. It would be easier to integrate in $\mathbb{R}$ by looking at “events” $B \in \mathcal{B}$ and applying typical integration methods (Fundamental Theorem of Calculus). Do we have an equivalent measure we can use for that purpose that yields the same results?

- For this we need two things: the **Radon-Nikodým Theorem** and the **Change of Variables Formula**. These two tools will allow us to introduce the definition of distributions and to safely switch spaces and measures when computing expectations.
- But first we need some (abstract) definitions.

Relation to Measure Theory

The Radon-Nikodým Theorem: Setup

- **Definition 3.9:** A measurable measure space is said to be σ -**finite** whenever Ω can be written as a countable union of measurable sets with finite measure. (Any finite μ).
- **Definition 3.10:** A measure ν is said to be **Absolutely Continuous** with respect to another measure μ , written $\nu \ll \mu$, whenever

$$\forall A \in \mathcal{F}, \mu(A) = 0 \implies \nu(A) = 0. \quad \text{Examples.}$$

- To motivate the theorem, let μ be a measure and let $\phi \geq 0$ be a measurable function from $(\Omega, \mathcal{F}, \mathbb{P})$ to $(\mathbb{R}, \mathcal{B})$. Now define a new function on $\mathcal{F}$:

$$\nu(A) = \int_A \phi \, d\mu = \int_{\Omega} \phi \mathbb{1}_A \, d\mu \quad \forall A \in \mathcal{F}$$

This is actually a new measure, e.g., if $\phi(\omega) = 1$ then $\nu(A) = \mu(A)$. So you can view ϕ as a function that will “deform” the measure μ and will end up being another measure.

Relation to Measure Theory

The Radon-Nikodým Theorem: Setup

- **Exercise: Prove that $\nu(A)$ above is a measure.** (*Hint: Use the definition of measure*).
- We thus know that given μ , ϕ , and A , we can integrate and get a new measure ν .
- The **Radon-Nikodým Theorem** tries to answer: given μ , A , and ν , can we find $\phi \geq 0$ that satisfies:

$$\nu(A) = \int_A \phi \, d\mu$$

Which is way more difficult! And the theorem gives both existence and uniqueness!

- The RN Theorem says that an absolute continuity condition is **necessary and sufficient** for the existence and uniqueness of $\phi \geq 0$.

Relation to Measure Theory

The Radon-Nikodým Theorem

Theorem 3.2 (Radon-Nikodým Theorem)

Let μ and ν be two σ -finite measures such that $\nu \ll \mu$. Then,

1. There exists a positive measurable function $\phi : (\Omega, \mathcal{F}) \longrightarrow (\mathbb{R}, \mathcal{B})$ such that:

$$\nu(A) = \int_A \phi \, d\mu \quad \forall A \in \mathcal{F}$$

The function ϕ is written $\phi = d\nu/d\mu$ and is called the Radon-Nikodým derivative of measure ν with respect to μ .

2. The function ϕ is unique almost everywhere μ (a.e.- μ) (i.e., all the sets in which they differ have measure μ zero).

We will not prove the theorem.

- How does this help with expectations? We can now define the concept of distribution.

Relation to Measure Theory

Induced Measure and Distribution

- Recall that $X : (\Omega, \mathcal{F}, \mathbb{P}) \longrightarrow (\mathbb{R}, \mathcal{B})$. Even though $\mathbb{P}$ is too complex, we can define a probability measure $\nu_X(B)$ on $\mathcal{B}$.
- **Definition 3.11:** For a random variable X , define the measure

$$\nu_X(B) = \mathbb{P}(X \in B) = \int_{\Omega} \mathbb{1}_{\{\omega \in X^{-1}(B)\}} d\mathbb{P} \quad \forall B \in \mathcal{B}$$

This is called **Induced Measure** in measure theory and **Distribution** in probability theory. $\nu_X(B) : (\mathbb{R}, \mathcal{B}) \longrightarrow [0, 1]$.
 $x \mapsto p$

- If we evaluate the distribution $\nu_X(B)$ at the Borel set $B = (-\infty, x]$, we get the **CDF** we saw before (check this). This definition is thus much more general and abstract.
- Instead of looking at probabilities $\mathbb{P}$ of events, we look at Borel sets $B \in \mathcal{B}$ and then look X^{-1} , which takes us back to $\mathcal{F}$. We can thus use it for integration: Change of Variables Formula.

Relation to Measure Theory

The Change of Variables Formula

Theorem 3.3 (Change of Variables Formula)

Let $X : (\Omega, \mathcal{F}, \mathbb{P}) \longrightarrow (\mathbb{R}, \mathcal{B})$ and let $h : (\mathbb{R}, \mathcal{B}) \longrightarrow (\mathbb{R}, \mathcal{B})$ be measurable and integrable. Then,

$$\mathbb{E}[h(X)] = \int_{\mathbb{R}} h \, d\nu_X = \int_{\mathbb{R}} h(x) \, d\nu_X(x)$$

Proof.

- We can compute probabilities of real-valued events, i.e., $\mathbb{P}(X \in B)$, and the expectation of a random variable, i.e., $\mathbb{E}[X]$, just by taking $h(X) = \mathbb{1}_B$ and $h(X) = X$.
- We can now construct PDFs and PMFs. We can do this by checking an absolute continuity condition and applying the Radon-Nykodým Theorem. This condition will be slightly different for continuous and discrete random variables.

Relation to Measure Theory

PDFs and PMFs

- A random variable X is said to be **continuously distributed** whenever its distribution ν_X is **absolutely continuous with respect to the Lebesgue measure** λ . Since λ is σ -finite, the RN Theorem applies and gives the existence and uniqueness of a measurable function $\phi(x)$ such that $d\nu_X = \phi d\lambda$. This function is the **PDF** of X . Hence:

$$\mathbb{E}[h(X)] = \int_{\mathbb{R}} h d\nu_X = \int_{\mathbb{R}} h \phi d\lambda = \int_{\mathbb{R}} h(x)\phi(x) dx$$

- A random variable is said to be **discrete** whenever its range S is either finite or countable, in which case its distribution is **absolutely continuous with respect to the counting measure** on S , i.e., $\nu_X \ll \mu_S$ where $\mu_A = \text{card}(A)$ for all $A \subseteq S$. By the RN Theorem, there exists a unique non-negative measurable function $\phi(x)$ such that $d\nu_X = \phi d\mu_S$. This function is the **PMF** of X . Hence:

$$\mathbb{E}[h(X)] = \int_{\mathbb{R}} h d\nu_X = \int_{\mathbb{R}} h \phi d\mu_S = \sum_{x \in S} h(x)\phi(x)$$

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